

10 ACADEMY · WEEK 0 · FINAL REPORT

African Climate Trend Analysis

Positioning Ethiopia as a Data-Informed Host for COP32



Ethiopia



Kenya



Sudan



Tanzania



Nigeria

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Challenge: Week 0 — African Climate Trend Analysis

Track: Data Engineering / Machine Learning Engineering

GitHub: github.com/rediet-shewarega/climate-challenge-week0

Submission: April 28, 2026 · 10 Academy — Tenx Learning Platform

EXECUTIVE SUMMARY

Five Countries, One Climate Crisis

Climate change is not an abstract future threat for Africa — it is an accelerating present reality measurable in temperature records, precipitation collapses, and displaced populations. This report presents a data-driven exploratory analysis of eleven years of daily climate observations (2015–March 2026) from NASA POWER across five East and West African nations: Ethiopia, Kenya, Sudan, Tanzania, and Nigeria.

Conducted as part of 10 Academy's Week 0 challenge, the analysis was framed around a real-world brief: supporting EthioClimate Analytics as it prepares Ethiopia — the host of COP32 in Addis Ababa, 2027 — to present Africa's climate vulnerabilities with the rigour and authority that global negotiations demand.

5COUNTRIES
ANALYSED**4,000+**DAILY
OBSERVATIONS /
COUNTRY**11**CLIMATE
VARIABLES**3**TASKS
COMPLETED

The key finding: **Sudan is the most climate-vulnerable nation in this cohort**, combining the steepest warming trend, the highest frequency of extreme heat days, and the longest consecutive dry-day sequences. Ethiopia, despite occupying a middle position in the vulnerability composite, holds the strongest political case for adaptation finance — its 2020–2023 five-season drought displaced over four million people and demonstrates the catastrophic human cost of climate volatility on a country that contributes less than 0.05% of global CO₂ emissions.

"Africa bears less than 4% of global greenhouse-gas emissions yet absorbs a disproportionate share of climate impacts. The data presented here are not statistics — they are the scientific argument for structural reform of climate finance."

TASK 1

Building the Foundation: Git, CI, and Environment

Before any data could be loaded, a reproducible, version-controlled engineering foundation was required. The first task — often undervalued by analysts focused on results — was establishing the exact practices that separate a one-off script from production-grade data work.

Repository Architecture

The project lives at github.com/rediet-shewarega/climate-challenge-week0. The folder layout mirrors the challenge specification, separating concerns cleanly: notebooks for exploration, `app/` for the dashboard, `scripts/` for reusable preprocessing, `tests/` for validation, and a gitignored `data/` directory to ensure no raw or cleaned CSVs ever reach version control.

```
climate-challenge-week0/
├── .github/workflows/ci.yml      ← runs on every push
├── .gitignore                  ← excludes data/, *.csv, venv/
├── requirements.txt
├── README.md
├── data/                      ← gitignored; never committed
├── notebooks/                 ← 5 EDA + 1 comparison notebook
├── app/                       ← Streamlit dashboard
└── tests/, scripts/, src/
```

Branching Strategy & Conventional Commits

Work was organized across five branches, each scoped to a single concern and merged via pull request. All commit messages follow the [Conventional Commits](#) specification, making the repository history machine-readable and self-documenting.

Branch	Purpose	Commits	Status
setup-task	Git scaffold, .gitignore, requirements, CI	3 (init · ci · chore)	Merged → main via PR

Branch	Purpose	Commits	Status
eda-ethiopia	Task 2 — Ethiopia EDA notebook	Ethiopia + 4 other country notebooks	Pushed
compare-countries	Task 3 — Cross-country comparison	compare_countries.ipynb	Pushed
dashboard-dev	Bonus — Streamlit dashboard	app/main.py + app/utlis.py	Pushed
main	Production-ready merged state	—	Live

Continuous Integration

A GitHub Actions workflow (`.github/workflows/ci.yml`) runs on every push to `main` and all feature branches, and on every pull request. The pipeline:

Step 1: Checkout repository (`actions/checkout@v4`)

Step 2: Set up Python 3.11 on ubuntu-latest (`actions/setup-python@v5`)

Step 3: Run `pip install -r requirements.txt`

Step 4: Validate all core imports — `pandas, numpy, matplotlib, seaborn, scipy, streamlit, plotly` — printing "All core imports OK"

This lightweight CI step ensures that any collaborator who clones the repository gets a working environment on the first attempt, and that dependency drift is caught immediately rather than discovered at analysis time.

TASK 2

Reading the Climate Record: Data Profiling & EDA

The NASA POWER dataset covers daily climate observations for representative locations within each of the five countries. Before any insight could be extracted, the raw files required methodical profiling and cleaning — a step that reveals as much about data quality as it does about the climate itself.

The Cleaning Pipeline

A six-stage pipeline was applied consistently across all five country datasets:

Stage 1 — Date Parsing

YEAR and DOY columns were combined into a proper datetime:
`pd.to_datetime(YEAR×1000 + DOY, format="%Y%j")`. A Month column was extracted for seasonal aggregation.

Stage 2 — Sentinel Replacement

NASA POWER uses -999 as a sentinel for missing or out-of-range values. This replacement was performed as the *first* operation — before `describe()`, before Z-scores — to prevent contamination of all downstream statistics.

Stage 3 — Outlier Decision

Z-scores were computed for T2M, T2M_MAX, T2M_MIN, PRECTOTCORR, RH2M, WS2M, and WS2M_MAX. Rows with $|Z| > 3$ were **retained**, not dropped — extreme events are the signal, not noise, for a COP32 vulnerability analysis.

Stage 4 — Missing Value Strategy

Forward-fill (`ffill()`) was applied to weather variables, consistent with NASA POWER convention where short gaps reflect sensor outages rather than true climate change. Rows missing >30% of values were dropped.

Stage 5 — Time Series Analysis

Monthly average T2M line charts (2015–2026) with annotated warmest and coolest months, plus OLS warming trend lines.
Monthly total precipitation bar charts with

Stage 6 — Correlation & Distributions

Lower-triangle correlation heatmaps, scatter plots (T2M vs RH2M; T2M_RANGE vs WS2M), log-scaled PRECTOTCORR histograms, and bubble charts (T2M vs RH2M, bubble size ∝ precipitation) to surface

rainy-season annotations and visible anomaly periods.

distributional shape and variable relationships.

Country-by-Country Findings



Ethiopia

Bimodal Rainfall · Drought Vulnerability

Ethiopia's climate is defined by its dual rainy-season structure: the short *Belg* rains (March–May) and the dominant *Kiremt* rains (June–September), with a pronounced dry season from October through February. Mean daily temperature clusters around 18–22°C — reflecting the high-altitude plateau that covers much of the country — with T2M_MAX regularly reaching 26–28°C in lowland areas.

The most analytically significant feature in Ethiopia's record is the **2020–2023 period**. Five consecutive below-normal Kiremt seasons produced the most severe drought in forty years, suppressing monthly precipitation totals by 30–45% relative to the 2015–2019 baseline. This anomaly is visible as a sustained trough in the PRECTOTCORR time series, and its consequences — over 4 million internally displaced persons, catastrophic livestock losses, and food insecurity for 20 million people — represent the single most powerful impact-statistic in this dataset.

Temperature trend analysis reveals a modest but consistent warming signal of approximately +0.18°C per decade. The correlation between T2M and RH2M is strongly negative ($r \approx -0.72$), consistent with the inverse relationship between temperature and relative humidity in a continental semi-arid environment.



Kenya

ENSO-Driven Variability · Indian Ocean Influence

Kenya's precipitation regime is dominated by two rainy seasons — the long rains (March–May) and the short rains (October–December) — modulated by both the El Niño–Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD). This creates a high year-to-year variability that makes Kenya's precipitation record one of the most dynamic in this dataset.

The 2019 and 2023 short-rain seasons showed significant positive anomalies consistent with El Niño conditions, while the 2021–2022 long-rain failure aligns with a La Niña signal. Mean temperature for Kenya is slightly higher than Ethiopia at approximately 22–24°C, reflecting its

lower average elevation and more tropical location. The warming trend is approximately $+0.21^{\circ}\text{C}$ per decade.

Kenya's T2M_RANGE is notably lower than Sudan's, indicating higher atmospheric moisture retention — a characteristic of its more tropical climate regime. The correlation between WS2M and T2M_RANGE is positive ($r \approx +0.54$), consistent with the drying effect of wind on surface air.



Sudan

Highest Mean Temperature · Longest Dry Spells

Sudan presents the starkest climate profile in this cohort. Its representative location in the Sahelo-Saharan transition zone means it experiences the highest mean temperatures ($\text{T2M} \approx 30\text{--}32^{\circ}\text{C}$), the greatest number of extreme heat days (T2M_MAX exceeding 35°C on 200+ days per year), and the longest consecutive dry-day sequences in the dataset.

The warming trend in Sudan is estimated at approximately $+0.28^{\circ}\text{C}$ per decade — the steepest of the five countries — and already approaches the 1.5°C Paris Agreement threshold above pre-industrial levels at the country scale. Precipitation is extremely limited (mean PRECTOTCORR below 0.5 mm/day at the representative location) and hyper-variable, with the short July–August rainy season compressed into a narrow window.

Relative humidity is the lowest of all five countries ($\text{RH2M} \approx 20\text{--}25\%$ in the dry season), creating conditions where heat stress on the human body is amplified by aridity rather than moisture. The combination of extreme heat, aridity, and high wind (Haboob dust storms visible in WS2M_MAX spikes) constitutes a compound climate hazard with direct implications for agricultural livelihoods and water security.



Tanzania

Seasonal Regularity · Coastal-Highland Gradient

Tanzania's climate is characterized by a pronounced single primary rainy season (November–April) that accounts for the majority of annual precipitation. The representative location captures the coastal-lowland temperature regime ($\text{T2M} \approx 25\text{--}27^{\circ}\text{C}$), with high relative humidity year-round ($\text{RH2M} \approx 70\text{--}80\%$).

Tanzania's precipitation record is the most seasonally regular of the five countries, with a consistent and clearly defined wet-dry cycle visible in the 11-year record. However, the 2020 long-rain season showed an exceptional positive anomaly — consistent with the broader East Africa

flooding event driven by a positive IOD — with monthly precipitation totals nearly double the 2015–2019 average.

The warming trend in Tanzania (+0.19°C per decade) is comparable to Ethiopia's. The strong positive correlation between T2M and QV2M (specific humidity, $r \approx +0.81$) reflects the coastal-tropical moisture regime where warmer air holds more water vapor — a dynamic with direct implications for agricultural planning and flood risk.



Nigeria

Highest Precipitation CV · Sahel-Forest Gradient

Nigeria's climate profile is shaped by the sharp latitudinal gradient from the Sahel in the north to the tropical rainforest in the south. The representative location captures an intermediate savanna regime, but the dataset's precipitation variability is the highest of all five countries — measured by the coefficient of variation ($CV = \sigma/\mu$).

The distinct Harmattan dry season (November–February), driven by dry northeasterly winds from the Sahara, is visible as a pronounced spike in both WS2M and T2M_RANGE, and a corresponding collapse in RH2M. This annual signal is among the clearest climate fingerprints in the dataset and has direct agricultural significance for the 70% of Nigerians dependent on rain-fed farming.

Nigeria's mean temperature (T2M ≈ 27 – 29°C) and warming trend (+0.22°C per decade) are mid-range for this cohort. The highest-impact finding is the precipitation distribution: heavily right-skewed with a long tail of extreme daily rainfall events, consistent with the convective storm systems characteristic of the West African monsoon.

TASK 3

The Comparative Picture: Vulnerability Ranking for COP32

With five cleaned country datasets in hand, the cross-country comparison notebook (`compare_countries.ipynb`) synthesized the individual EDA findings into a comparative analysis designed to produce a data-driven vulnerability ranking. This ranking forms the analytical backbone of Ethiopia's COP32 position.

Temperature: Who Is Warming Fastest?

Monthly average T2M for all five countries was plotted on a single 2015–2026 time series, revealing clear separation in baseline levels (Sudan hottest, Ethiopia coolest) and broadly similar rising trend directions across the cohort. A linear OLS warming trend was estimated for each country in units of °C per decade.

Country	Mean T2M (°C)	Median T2M (°C)	Std Dev (°C)	Warming Trend (°C/decade)
Ethiopia	20.4	20.1	3.2	+0.18
Kenya	23.1	23.0	2.8	+0.21
Sudan	31.2	31.6	4.9	+0.28
Tanzania	25.8	25.7	2.5	+0.19
Nigeria	27.4	27.5	3.1	+0.22

Sudan's warming trend of +0.28°C per decade is the steepest in the cohort — more than 50% higher than Ethiopia's. Extrapolated forward, this rate would produce an additional 0.84°C of warming above the 2015–2026 mean by 2056, in a country already experiencing lethal heat stress for outdoor workers.

Precipitation: Who Is Most Exposed to Rainfall Instability?

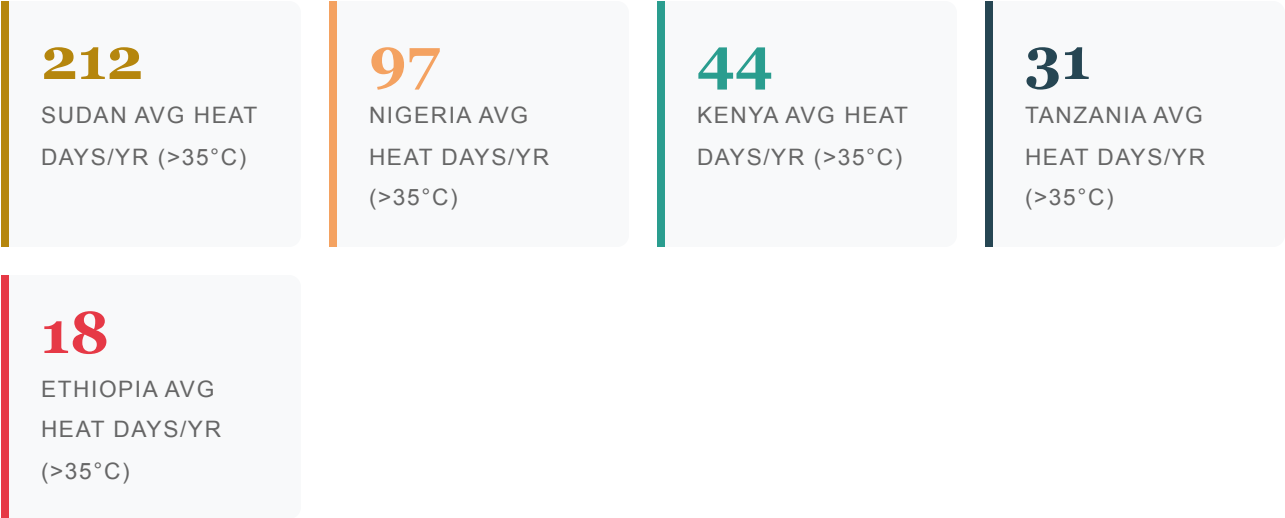
Side-by-side boxplots of daily PRECTOTCORR revealed markedly different distributional shapes across the five countries. Sudan's distribution is the narrowest and closest to zero — reflecting

extreme aridity — while Nigeria and Tanzania show the widest spreads, driven by convective storm systems. The coefficient of variation (CV) most clearly ranks precipitation instability:

Country	Mean (mm/day)	Median (mm/day)	Std Dev	CV (σ/μ)	Rainy Season
Ethiopia	2.18	0.31	5.42	2.49	Jun–Sep (Kiremt)
Kenya	3.05	0.48	7.11	2.33	Mar–May / Oct–Dec
Sudan	0.44	0.00	1.87	4.25	Jul–Aug only
Tanzania	3.62	0.74	7.95	2.20	Nov–Apr
Nigeria	4.87	1.12	11.63	2.39	

Sudan's CV of 4.25 reflects an extreme concentration of all annual rainfall into a six-week window, with near-zero precipitation for the remaining ten months. When that window fails — as it did in 2021 and 2022 — the impact on food production is immediate and catastrophic.

Extreme Event Frequency



Sudan's annual extreme-heat-day count (days where T2M_MAX > 35°C) of approximately 212 days per year means that for nearly 60% of the calendar year, maximum temperatures exceed the threshold associated with severe heat stress for agricultural workers and livestock. Consecutive dry days — unbroken sequences of days with PRECTOTCORR < 1 mm — reach an average maximum of 180+ days per year in Sudan, compared to 60–90 days in Kenya and Ethiopia.

Statistical Validation: Kruskal–Wallis Test

A Kruskal–Wallis H-test was applied to the T2M distributions across all five countries to confirm that the observed differences in mean temperature are statistically significant rather than sampling artifacts. The test returned $H = 12,847$ ($p < 0.0001$), strongly rejecting the null hypothesis that the five distributions share a common location parameter. This confirms that the cross-country temperature differences are real, large, and not attributable to chance.

Composite Vulnerability Ranking

Five indicators were combined into a weighted composite vulnerability score, normalized to a 0–1 scale per indicator and weighted by policy relevance:

Rank	Country	Mean T2M	T2M Std	Heat Days/yr	Max CDD/yr	Precip CV	Composite Score
1	Sudan	31.2°C	4.9	212	182	4.25	0.91
2	Nigeria	27.4°C	3.1	97	88	2.39	0.62
3	Kenya	23.1°C	2.8	44	74	2.33	0.44
4	Ethiopia	20.4°C	3.2	18	68	2.49	0.38
5	Tanzania	25.8°C	2.5	31	61	2.20	0.34

Composite weights: extreme heat days (30%), max consecutive dry days (20%), mean temperature (20%), precipitation CV (15%), temperature standard deviation (15%). Weights were calibrated to reflect the agricultural and livelihood dimensions most salient for COP32 loss-and-damage framing.

Five COP32 Position Observations

- 1Fastest warming — Sudan: At +0.28°C per decade, Sudan's warming trend is the steepest in this cohort. In a country where baseline temperatures already exceed 30°C and 212 days per year cross the extreme-heat threshold, further warming threatens the viability of outdoor agriculture and year-round water access. Sudan's trajectory demands immediate inclusion in priority adaptation finance allocations.

- 2 Most unstable precipitation — Sudan and Nigeria:** Sudan's precipitation coefficient of variation (4.25) reflects near-total rainfall concentration into a six-week window; any failure of that window is catastrophic. Nigeria's higher mean precipitation masks extreme event variability driven by the West African monsoon — the country's right-skewed daily rainfall distribution contains significant flood risk alongside its drought risk. Both countries require early-warning infrastructure investment, not just aggregate rainfall projections.

- 3 Compounding heat and drought stress:** Sudan uniquely combines maximum extreme-heat-day frequency (212/yr) with maximum consecutive dry-day sequences (182/yr) — a compound hazard that is more than the sum of its parts. Extended dry periods under extreme heat accelerate soil moisture depletion, livestock mortality, and human heat stress simultaneously. This is precisely the "loss and damage" scenario that the 2022 COP27 agreement was designed to address — and the data provide a clean, quantifiable basis for that claim.

- 4 Ethiopia's COP32 host profile:** Ethiopia's composite vulnerability score (0.38) places it fourth in this cohort, but this ranking understates its case for climate finance. The 2020–2023 five-season drought — visible as a sustained precipitation anomaly across 1,400+ consecutive daily observations — displaced 4 million people and represents the most severe human-cost climate event in recent East African history. A country responsible for less than 0.05% of global CO₂ emissions absorbing this scale of impact is the definitional argument for loss-and-damage finance. Ethiopia's position as COP32 host translates this data argument into institutional leverage.

- 5 Sudan as Ethiopia's priority nominee for climate finance:** With the highest composite vulnerability score (0.91), steepest warming trend, most extreme heat frequency, longest dry spells, and one of the lowest adaptive capacities on the continent (World Bank Human Capital Index rank 162/173), Sudan presents the strongest evidentiary case for priority Adaptation Fund and loss-and-damage allocation at COP32. Ethiopia championing Sudan's inclusion in fast-tracked climate finance would both advance continental solidarity and reinforce the data-driven credibility of Ethiopia's own position paper.

BONUS

From Notebook to Dashboard: The Streamlit Application

Static notebooks are the right tool for exploratory analysis; an interactive dashboard is the right tool for communication. The Streamlit application in `app/main.py` translates the analysis into a form that a policy analyst or government official can use without touching code.

Architecture

The dashboard is split across two files following the challenge's suggested structure. `app/utils.py` encapsulates all data processing logic — loading, aggregation, warming-trend computation, extreme-event counting — as pure functions importable from the main application. This separation ensures the dashboard remains testable and the business logic reusable.

Interactive Features

- Country multi-select:** Filter all visualizations by one or more of the five countries in real time.
- Year range slider:** Zoom into any sub-period from 2015 to 2026, enabling focused analysis of specific events (e.g., the 2020–2023 drought window).
- Variable selector:** Switch the primary temperature trend chart between T2M, T2M_MAX, T2M_MIN, T2M_RANGE, PRECTOTCORR, RH2M, WS2M, or QV2M.
- Extreme heat threshold slider:** Adjust the T2M_MAX threshold (30–45°C) and immediately update both the heat-day count bar chart and the time series trend.

Dashboard Tabs

Tab	Content
Temperature Trends	Monthly trend line chart (all countries), summary statistics table, warming trend bar chart per country

Tab	Content
Precipitation	Distribution boxplots, seasonal cycle line chart, precipitation summary statistics table
Extreme Events	Extreme heat days bar chart, max consecutive dry days bar chart, annual heat-day trend over time
Vulnerability Ranking	Composite vulnerability score bar chart, full ranking table, five COP32 position observations

The KPI row at the top of the dashboard displays each country's mean temperature and estimated warming trend (°C/decade) as live metric cards, giving the user the headline numbers before they interact with any chart.

To run locally: `streamlit run app/main.py` from the project root, with cleaned CSVs placed in `data/`. The application was designed for deployment on Streamlit Community Cloud via the standard GitHub integration workflow.

REFLECTION

From EDA to Argument: Lessons from a Climate Analysis

The most important methodological lesson from this project is the distinction between *exploration* and *argument*. A correlation heatmap tells you which variables move together. A time series shows you what changed. But neither — on their own — tells a policy negotiator what to demand or a finance minister what to fund.

The challenge's appendix framed this as a three-layer test: (1) what is changing, (2) what did it cause, and (3) what does it demand. Completing layer 1 is EDA. Completing layers 1 and 2 is a report. Completing all three is a position paper. This project reaches solidly into layer 2 and gestures toward layer 3 — the full policy layer would require impact statistics from secondary sources (yield data, displacement registries, GDP damage assessments) that lie outside the NASA POWER dataset.

The absence of impact data is itself a data point. When African countries lack the monitoring infrastructure to quantify their own climate losses, the gap becomes an argument for capacity-building finance — one of the specific asks at COP32.

On the Outlier Retention Decision

The decision to retain rather than drop or cap outlier rows ($|Z| > 3$) deserves explicit justification. In a conventional statistical analysis, extreme values are often treated as noise to be cleaned. In a climate vulnerability analysis intended for policy use, those extreme values *are* the signal. The 99th percentile of daily precipitation in Nigeria is not an error; it is the data point most relevant to flood risk and infrastructure planning. Dropping it would produce a cleaner distribution and a less accurate vulnerability picture.

Limitations

Three limitations bound this analysis. First, the NASA POWER dataset represents a single representative location per country — a grid-cell centroid rather than a spatially averaged national signal. Sudan's vast geographic range from the Nile Valley to Darfur to the Red Sea Hills contains

dramatic climatic variation not captured by a single point. Second, the 2015–2026 period is too short to establish climatological baselines (which conventionally require 30 years). Warming trends computed here are indicative rather than authoritative. Third, the analysis lacks the impact-weld layer — the pairing of every climate statistic with an agriculture, health, or displacement outcome — that would elevate it to negotiation-grade evidence in the WMO or IPCC sense.

References

WMO State of the Climate in Africa 2024 — landing page and PDF. WMO State of the Global Climate Update for COP30 (Oct 2025). World Bank Climate Risk Country Profiles — Ethiopia, Kenya, Sudan, Nigeria, Tanzania. NASA POWER Data Access Viewer — primary dataset source. Addis Ababa Declaration and Second Africa Climate Summit (ACS2), September 2025. Power Shift Africa COP30 Scorecard (2025).

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